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LeafyResNet: Fusarium wilt detection in lettuce using UAV RGB imaging and advanced deep learning model

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ABSTRACT

Lettuce, one of the most consumed leafy greens globally, offers significant health benefits due to its high vitamin, mineral, and fiber content. However, Fusarium wilt, a soil-borne fungus, threatens lettuce yields by reducing both quality and quantity. Traditional disease detection methods, such as manual inspection, are time-consuming and inefficient. This study proposes an Unmanned Aerial Vehicle (UAV)-based- approach for detecting Fusarium wilt in lettuce using high-resolution Red-Green-Blue (RGB) imagery. (1) a high resolution RGB lettuce dataset captured by drones at approximately 10 m altitude in collaboration with the Yuma Center of Excellence for Desert Agriculture (YCEDA), (2) identification of candidate Fusarium-infected regions by evaluating 300x300 pixel image patches for light tan coloration, followed by the application of a customized Residual Neural Network (ResNet), called LeafyResNet, to confirm Fusarium presence, and (3) a method for quantifying Fusarium infection severity, which was validated against an expert-ground truth. In this study, 6,239 high-resolution RGB images (4000 × 3000 pixels) were collected over nine weeks from YCEDA fields and cropped into 675 ROI patches (100 × 100 pixels) representing healthy and Fusarium-infected areas. Data augmentation expanded the training set to 57,000 patches, while the testing set (20%) remained unaugmented to ensure fair evaluation on unseen data. Our approach to detect Fusarium wilt achieves 96.30% accuracy, 94.10% precision, 100% recall, 99.5% mean Average Precision (mAP) and a 97.10% F1-score, with a 4% false positive rate. Disease severity scores showed an overall accuracy of 86%. LeafyResNet has been specifically adapted with ResNet as a base. Overal, the adapted LeafyResNet performed better than the best baseline, ResNet34, with a 0.3% higher accuracy and 0.8% higher F1-score, showing measurable improvements over ResNet18, ResNet34, Inception_v3, and VGG16, and highlighting the benefits of customizing models for agricultural applications. Core of the adaptation is the starting layers and residual blocks, which tailored LeafyResNet to capture subtle wilting symptoms in leaves as multi-scale features in UAV datasets, providing an efficient and scalable solution for precision agriculture.

1. Introduction

Lettuce (*Lactuca sativa*), a globally consumed leafy vegetable, is valued for its low calorie content and rich nutritional profile, including fiber, iron, folate, and vitamin C [33]. In the United States, it represented nearly 20% of the $21.8 billion vegetable and melon market in 2022 [57]. However, lettuce production is increasingly challenged by factors such as water scarcity, pest pressure, and soil nutrient depletion [55], all of which contribute to crop vulnerability. One significant threat is *Fusarium oxysporum* f. sp. *lactucae*, the causal agent of Fusarium wilt, which severely affects lettuce health and yield.

Using resistant lettuce varieties is still the most common and effective way to manage Fusarium wilt. However, only a few resistant cultivars are available [13], and most are suitable for early planting seasons (e.g., planting in September to early October, harvesting in late November to December) in warmer conditions. Knowing which areas are at risk helps to improve planning. In recent years, the disease has been showing up during all planting windows, increasing the need for additional support tools such as detection technologies. Therefore, early

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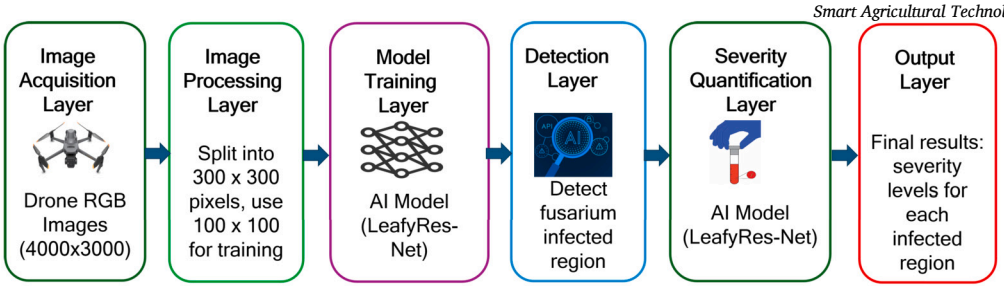

Fig. 1. Layered architecture for Fusarium wilt detection in lettuce using drone RGB images. The diagram illustrates the flow of data through the various stages: from image acquisition, preprocessing, and model training, to detecting infected regions, quantifying infection severity, and generating final severity labels.

disease detection not only helps farmers respond faster, but also makes disease control more efficient and cost-effective.

Traditional disease detection approaches — primarily reliant on manual inspection — are labor-intensive and often reactive, resulting in widespread infection before mitigation can be applied [54].

Traditional disease detection approaches are labor-intensive and time-consuming [54]: obtaining conclusive results can take days to weeks due to the time required for organism growth and confirmatory assays [36,15,40]. Detection of viral pathogens using indicator plants may also take several weeks depending on the plant species [37,46]. Previous studies on Fusarium wilt in radish have noted that manual inspection is often time-consuming and unproductive [8]. Although chemicals are not commonly used to treat Fusarium wilt, they can be applied in small sections where the disease is found. For example, chloropicrin fumigation is too expensive to treat a whole field, but it may be practical for smaller, high-risk areas prior to planting [60]. Other methods like solarization, bio-solarization, and steam are also more affordable when used in small, targeted spots [20]. Therefore, early detection allows farmers to find these zones and apply treatments only where necessary, reducing costs and limiting the spread of the disease. It also helps identify which parts of a field have higher levels of disease-causing soil inoculum [20], allowing growers to choose suitable lettuce types for the next season. Since fields are often planted with several varieties depending on market demand, knowing where the disease is likely to help farmers plan where to plant each type. Lastly, if severe infection is expected in a certain part of the field, farmers can reduce inputs such as water, fertilizer, and pest control in that area, or avoid sending equipment through it, both of which can help limit the spread of disease to healthy plants. Consequently, there is a critical need for scalable, accurate, and early-stage detection methods [7].

Recent advancements in remote sensing technologies [18,17] have facilitated rapid, non-invasive disease monitoring in agriculture [14,26,10,5,58]. Low-altitude platforms, such as Unmanned Aerial Vehicles (UAVs), offer an affordable and precise means of acquiring high-resolution imagery for crop analysis [28]. Combined with spectral imaging and machine learning, UAVs enable real-time assessment of plant health and early disease detection [8,39,3]. Detecting Fusarium wilt in lettuce via aerial imagery presents unique challenges. The disease often manifests as subtle yellow-tan discoloration closely resembling the surrounding soil, complicating traditional image-based classification [59]. Previous efforts using conditional generative adversarial networks (cGANs) [29] for soil segmentation [41] and U-Net architectures [34] for Fusarium wilt detection [56] have struggled with misclassifying infected plants as soil due to color similarity. This problem is particularly relevant in the Sonoran Desert near Yuma, Arizona, which produces approximately 90% of U.S. lettuce from November through April. Here, the soil often shares visual properties with symptomatic leaves—particularly in the light tan color range (RGB: 225, 220, 160).

To address this, we trained a deep learning model to emphasize variation within this light tan color range, to improve the discrimination of infected plant tissue from background soil. Although YOLO and its mini-variants are widely used for object detection in agricultural setting [1] and UAV based monitoring [35]. YOLO is preferred when fast detection and approximate feature areas marked by bounding boxes are sufficient. However, we chose ResNet as a foundation because its accuracy and clear feature representation are application requirements, especially on smaller datasets that are practical to be collected by farmers. Therefore, we introduce LeafyResNet, a deep learning framework for accurate detection and severity estimation of Fusarium wilt in lettuce using UAV-acquired RGB imagery.

Over the last two years several studies show that CNN-based models remain very effective for crop disease detection. For example, Majeed et al. [44] used GoogleNet and MobileNetV2 for tomato disease detection on edge devices, and Aishwarya and Reddy [49] achieved convincing results with DenseNet169, Inception, and Xception for groundnut leaf classification. Leite et al. [38] reviewed CNN approaches for disease severity assessment, Mengesha and Mengistie [47] used MobileNetV3, InceptionV3, and DenseNet201 for tomato leaf detection with explainable AI, and Katumba et al. [30] developed BeanWatchNet, a CNN-based model, for real-time detection of bean plant diseases. These studies show that the accuracy of classifications by CNN models, including ResNet, is important for screening of plant diseases.

Earlier UAV studies have looked at Fusarium wilt detection in radish [9], Yet, no work has focused on UAV- based detection in lettuce, especially in Arizona desert soils, where the soil often shows a similar color profile than symptomatic leaves. To overcome this challenge, our method uses UAV images collected from Arizona desert fields and divides each image into 300×300 pixel patches. This allows the model to focus on fine-grained patterns and subtle color differences that indicate early disease symptoms. By leveraging the strengths of ResNet's residual learning architecture [24,53], LeafyResNet effectively learns complex, nonlinear features from localized image data. Additionally, the model estimates a Fusarium Infection Severity Index (FISI), which quantifies disease severity on a scale from 0 (healthy) to 4 (severely infected), based on expert-labeled data from the Yuma Center of Excellence for Desert Agriculture. Fig. 1 shows the layered architecture of the Fusarium wilt detection model for lettuce on an example of a collected drone RGB image. The diagram illustrates the sequential flow of data across different stages, starting from the acquisition of raw images, through preprocessing, model training, and region detection, to the quantification of Fusarium infection severity and the final output of severity labels. This architecture outlines the key processes for automated detection and severity assessment of Fusarium wilt in crops using AI-based methods. The key contributions of this work include:

- **The first automated method** for detecting Fusarium wilt in lettuce using UAV-acquired imagery, addressing the gap of lettuce-specific UAV disease monitoring.
- **Light-tan intensity analysis** that improves early detection of Fusarium wilt by distinguishing infected tissue from visually similar soil, overcoming misclassification challenges in traditional image-based methods.
- **Performance evaluation of LeafyResNet** against established architectures (ResNet18, ResNet34, VGG16, Inception_v3), shows increased accuracy in detection and severity quantification of Fusarium wilt in lettuce.

Table 1
Details of the RGB lettuce dataset collected in Yuma County, Yuma, Arizona, using the DJI Inspire 1 drone at an altitude of 10 meters. The dataset includes images from two adjacent fields: the North field with 2,704 images and the South field with 3,535 images, totaling 6,239 images.

Region (Yuma County, Arizona)	Date	#Fields	Altitude (m)	#RGB Images
North field	Oct 11 - Dec 13, 2023	1	10	2,704
South field	Oct 11 - Dec 13, 2023	1	10	3,535
Total		2	10	6,239

- **A labeling approach** that assigns severity scores to infected regions, enabling detailed monitoring of disease progression; no prior UAV-based AI method exists for lettuce.

The remainder of this paper is organized as follows. Section 2 details the lettuce dataset and preprocessing pipeline. Section 3 describes the patch-based detection framework, including the development of the proposed LeafyResNet model and FISI estimation method. Section 4 presents experimental results, model evaluations, and a comparison with existing approaches. Section 5 concludes the study and outlines directions for future research.

2. Lettuce dataset description

2.1. Study area, cultivars, and data acquisition

Lettuce is grown in Arizona's desert environment during the winter and spring months from September to April. Annually, Arizona lettuce growers lose approximately $9 million due to Fusarium wilt [12]. Lettuce planted in September that matures in November and December is at the highest risk of being infected with Fusarium. Therefore, in this study, lettuce was planted on September 19. The average air temperatures during the season were 29 °C in September, 25 °C in October, and 18 °C in November. The average soil temperatures at a depth of 100 mm were 34 °C, 29 °C, and 18 °C, respectively [4]. Fusarium wilt symptoms were observed and confirmed in the field during this study period.

Lettuce plants can be affected by different races of Fusarium wilt. However, currently, race one is the only known race present in the United States [50]. Although a new variant has been found in California that might be classified as a new race, it has not been detected in Arizona [50]. Tests from Arizona, including samples collected from our study field, have confirmed that race one is the cause here. Our study was performed in Yuma, AZ. We planted the cultivars 'El Guapo' (Syngenta), a susceptible cultivar; 'Tamarack' (3 Star Lettuce), a tolerant cultivar; and 'Desert Eagle' (Vilmorin), which is highly tolerant in a field with a silty clay loam soil type. The field is known to be naturally infested with *Fusarium oxysporum f.sp. lactucae*. Cultivars were direct seeded in 100-foot sections in two seed lines on 34-inch beds, then sprinkler-irrigated until germination. Two weeks after planting, the beds were thinned to a 12 inch spacing between plants.

Data for this study were collected using a DJI Inspire 1 drone equipped with an RGB camera in a 2-acre field in Yuma County, Arizona. Flights were carried out by trained staff at the Yuma Center of Excellence for Desert Agriculture. A total of 6,239 images were obtained from two adjacent fields: the north field (2,704 images) and south field (3,535 images). The data collection spanned nine weeks, covering two regions within the field, with a total of 6,239 images captured from lettuce plants of various genotypes. The Yuma fields used in our study were only affected by Fusarium wilt.

Data acquisition began approximately one month after planting, with weekly drone flights from October 11 to December 13, 2023. No data was collected during the seventh week due to adverse weather. The drone captured high-resolution images from an altitude of approximately 10 meters with a resolution of 4000 x 3000 pixels. In this study, a drone flight altitude of 10 meters was chosen for Fusarium wilt detection. This altitude falls within the recommended range of 10 to 20 meters [45], which offers higher spatial resolution and more precise detection of disease symptoms. However, it still provides a sufficient coverage area for efficient data collection. Our chosen flight is supported by previous research, which has shown that a 10-meter altitude yields better performance and higher positional accuracy compared to higher flight heights [2]. Moreover, images taken at this altitude have sufficient resolution to detect wilting while still covering a larger portion of the field. Details of the RGB lettuce dataset are summarized in Table 1.

Fusarium wilt, initially absent, began spreading in early November. By November 2, 10.23% of images were affected, with a sharp increase in the north field to 19.76% by November 7. Manual inspection found that approximately 44.85% of the images in both fields showed at least one wilting plant after nine weeks. An example of a UAV (drone) image of the lettuce field with highlighted Fusarium wilt symptoms is shown in Fig. 2.

2.2. Training dataset preparation

For training the LeafyResNet model, the dataset first needed to be prepared. A total of 30 full RGB images were selected from the north field, from which 675 image patches (100 × 100 pixels) were extracted, representing both healthy and Fusarium-infected regions. These Region of Interest (ROI) images were preprocessed to ensure consistency and improve model performance. Specifically, pixel values were normalized by scaling the pixel intensities to the range [0, 1], which helps the model converge faster during training.

The ROI patches (see Fig. 3) were then split into an 80:20 ratio for training and testing, with the training set containing 540 ROI images and the testing set containing 135 images. Data augmentation was applied only to the training set, expanding it to 57,200 image patches, while the testing set remained non-augmented to ensure a fair evaluation of the model on original, unseen data.

The data augmentation techniques used in this study included rotation, translation, shearing, zooming, and horizontal flipping. Rotation and flipping [27] were applied to introduce variability in image orientation. Shearing and zooming [21] were used to simulate distortions and scale variations, while translation [11] was implemented to shift images spatially, creating a more diverse dataset and improving model generalization.

After augmentation, the LeafyResNet model was trained to classify two categories: healthy and Fusarium wilt, confirming the presence of Fusarium or healthy tissue in each region.

3. Methodology

3.1. Patch creation

The patch-based image method to detect Fusarium wilt in lettuce is inspired by techniques used in agricultural image analysis. Previous work by Hasan et al. [23] on crop and weed recognition through patch-based deep learning, and by Moazzam et al. [48] on weed detection in sugar beet using patch-image classification demonstrates improved accuracy using patches for localized feature analysis. This segmentation enhances the model's ability to discern finer details within an image, a crucial aspect in plant disease detection, as small, localized changes in texture or color are significant indicators of disease. In addition, our

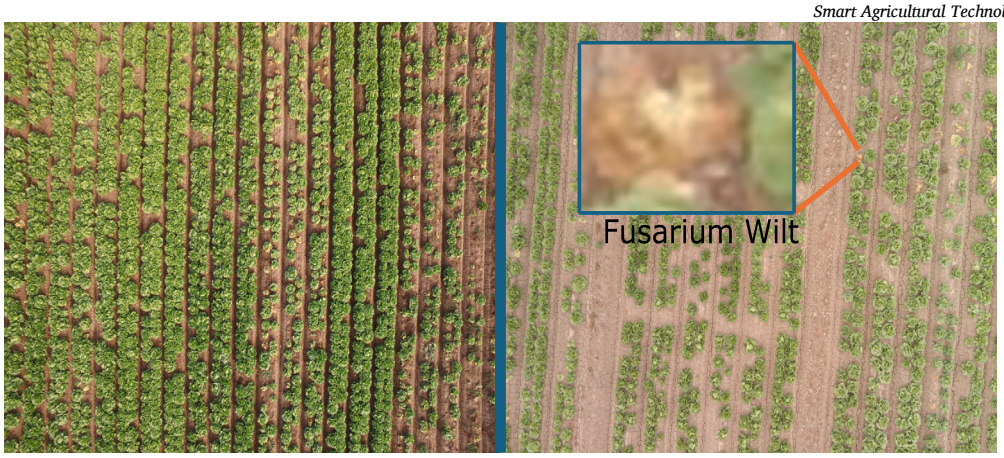

Fig. 2. UAV-captured RGB image and highlighted Fusarium wilt symptoms in lettuce field. Left: A high-resolution RGB image captured by a DJI Inspire 1 drone at an altitude of approximately 10 meters, showing the overall lettuce field. Right: A full image of the lettuce with Fusarium wilt symptoms, where areas affected by the disease are highlighted for clarity. The highlighted areas emphasize the typical yellowing and wilting of lettuce leaves due to Fusarium wilt, as observed during the study.

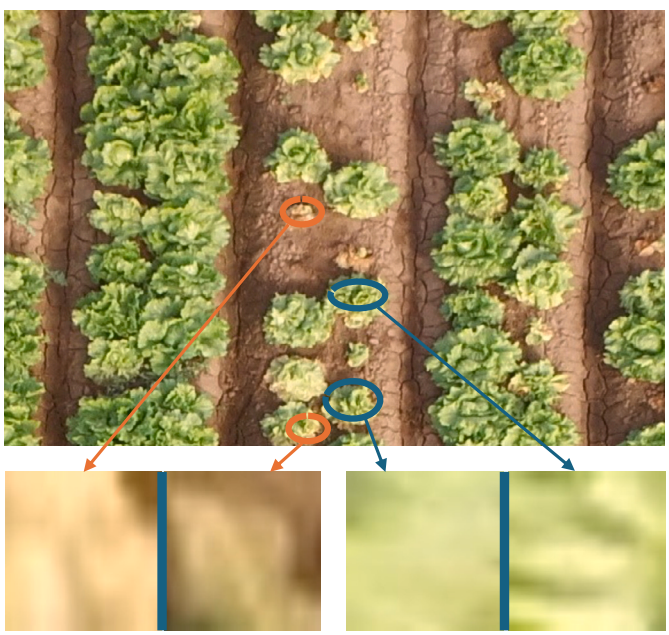

Fig. 3. An RGB image of a lettuce sample captured by UAV, showing two distinct ROI types: blue circles indicate healthy lettuce regions, while orange circles mark Fusarium-infected ROIs.

patch-based approach for localized Fusarium wilt detection is supported by similar multi-scale feature extraction methods applied in environmental monitoring, such as spectral morphology analysis of karst wetland water quality parameters [16].

To implement this localized analysis for the detection of Fusarium wilt, our method divides the original image, which has a resolution of 4000 × 3000 pixels (4K resolution), into smaller patches of 300 × 300 pixels. This patch size (300 × 300) was chosen to balance spatial resolution and computational efficiency: each patch is large enough to capture meaningful leaf features, including detailed differences between infected and healthy regions, while providing a sufficiently broad view for detection. Smaller patches would show only a tiny area at a time, offering limited additional information, whereas larger patches would reduce sensitivity to localized infection symptoms and increase computational complexity. Therefore, this size was selected after testing multiple patch dimensions and visually evaluating the distinction between infected and healthy regions.

For an original image with dimensions $H \times W$ (height × width) and a patch size of $S_x \times S_y$ (300 × 300 pixels), the total number of patches T_x and T_y along the width and height are calculated as:

$$T_x = \frac{W}{S_x}, \quad T_y = \frac{H}{S_y} \tag{1}$$

The image will then be divided into a grid of 13×10 patches, as the original 4000×3000 image is divided by the 300×300 patch size ($4000/300 = 13$ and $3000/300 = 10$). Each patch $p_{i,j}$ is extracted from the image starting at coordinates (x_0, y_0), where:

$$x_0 = i \times S_x, \quad y_0 = j \times S_y \tag{2}$$

S_x and S_y are set to 300, making each patch 300 × 300 pixels. These patches are used to estimate the light-tan color range, which aids in the identification of the candidate Fusarium infection.

3.2. Fusarium wilt detection using deep learning

3.2.1. Detection of Fusarium infected candidate regions

We evaluate the light tan color value as a feature for candidate Fusarium wilt detection. The process begins by evaluating the image patches created earlier for further processing. We sampled 50 image patches for soil color samples and generated a histogram. The median values of the histogram were calculated as R = 225, G = 208, and B = 160, which fall within the tan color range. Since Fusarium appears lighter than these values, we selected candidate pixels of Fusarium wilt as equal to or greater than the median values. After that, a mask is generated to isolate candidate regions of connected pixels. To further refine the mask and improve the detection of contours [51], image processing techniques are applied. Morphological operations, including dilation and erosion, are used to improve the precision of contour identification. Dilation expands the detected regions, merging neighboring areas that may have been separated because of noise or inconsistencies in the image. Erosion is then applied to reduce the size of the expanded regions, eliminating small, irrelevant features, and sharpening the boundaries of the contours. (Fig. 4.)

Finally, the contours are extracted from the processed image, to identify candidate areas of Fusarium wilt. This approach leverages the distinctive yellow-brown discoloration of infected plants, as well as the potential to efficiently identify Fusarium from soil color, which may resemble the color of the disease. Using contour detection, a rectangle box is created around each candidate region.

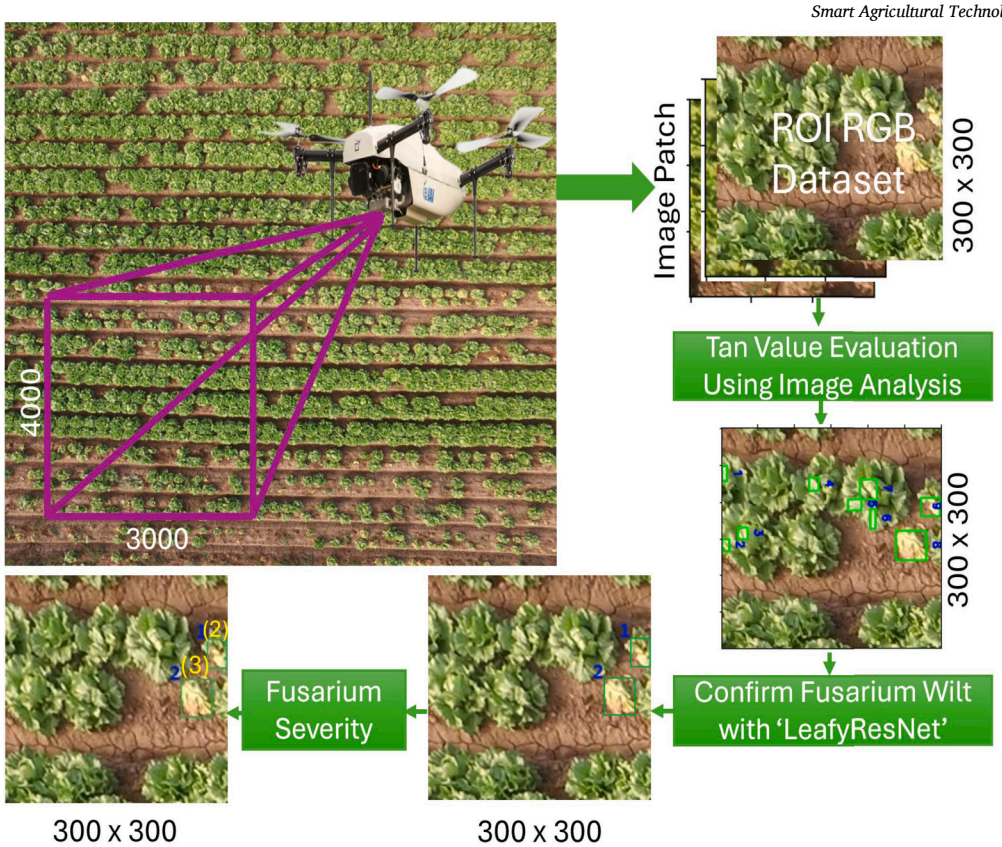

Fig. 4. Proposed methodology for Fusarium wilt detection in lettuce using UAV-based RGB imagery. Steps include Region of Interest (ROI) RGB dataset preparation from the full image, light tan intensity evaluation, classification with the LeafyResNet model, and severity quantification by labeling for effective agricultural monitoring.

Table 2
Layer-wise Architecture of ResNet for Fusarium Confirmation in Lettuce Crops.

Layer Type	Output Shape (Size)	Description
Input Layer	(100, 100, 3)	Input image with 3 color channels (RGB)
Conv1	(100, 100, 64)	7x7 convolution, 64 filters, stride = 2, BatchNorm, ReLU
MaxPooling	(50, 50, 64)	MaxPooling (3x3), stride = 2 to reduce spatial dimensions
Residual Block 1	(50, 50, 64)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 2	(50, 50, 64)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 3	(50, 50, 64)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 4	(25, 25, 128)	2x3x3 convolution, BatchNorm, ReLU, stride = 2, skip connection
Residual Block 5	(25, 25, 128)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 6	(25, 25, 128)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 7	(13, 13, 256)	2x3x3 convolution, BatchNorm, ReLU, stride = 2, skip connection
Residual Block 8	(13, 13, 256)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 9	(13, 13, 256)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 10	(7, 7, 512)	2x3x3 convolution, BatchNorm, ReLU, stride = 2, skip connection
Residual Block 11	(7, 7, 512)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Residual Block 12	(7, 7, 512)	2x3x3 convolution, BatchNorm, ReLU, skip connection
Global Average Pooling	(1, 1, 512)	Reduces the spatial dimensions to 1x1
Dense (Output)	(1,)	Fully connected layer, sigmoid activation for binary classification

3.2.2. Implementation of the LeafyResNet model

We used LeafyResNet to confirm candidate regions of Fusarium wilt. The model begins with an initial convolutional layer using a 7x7 kernel and 64 filters, followed by batch normalization and a ReLU activation function. A max-pooling layer with a 3x3 kernel and stride 2 is then applied to reduce the spatial dimensions. The core of the model consists of several residual blocks. Each block includes two convolutional layers with a 3x3 kernel, batch normalization, and ReLU activations. The residual connections within each block add the input to the output of the convolutions, helping the model learn residual mappings and mitigating the problem of vanishing gradients [25], thereby improving training efficiency.

LeafyResNet retains these core residual learning components of ResNet18 but is adapted specifically to detect Fusarium wilt in lettuce. The early convolutional layers are designed to enable the network to learn subtle light-tan variations between symptomatic leaves and soil, helping distinguish infected tissue from the background—a challenge not addressed in standard ResNet. In addition, LeafyResNet incorporates a few extra convolutional layers and modifies the configuration of some residual blocks to better capture multi-scale features, making it more effective for small and complex UAV datasets. Finally, the output layer is redesigned as a single sigmoid neuron to classify whether a patch contains disease. The architecture of LeafyResNet is detailed in Table 2.

3.3. Estimation of Fusarium Infection Severity Index (FISI)

Following the confirmation of Fusarium wilt regions by the LeafyResNet model, we hypothesize that the brightness of the pixels in the identified regions can be used to estimate FISI in the detected area. Specifically, we assume that regions with brighter pixels correspond to a more severe infection. To estimate infection severity, we perform a pixel-level analysis by converting the identified regions from RGB to the HSV color space, where H represents the Hue, S represents the Saturation, and V represents the Value, which is the brightness component.

The use of HSV for this analysis is motivated by its ability to separate image intensity from color information, making it more robust to lighting variations and shadow effects compared to the RGB color space. This approach has already been validated [8]. Here, the HSV color space was effective to analyze the severity of plant disease, particularly in outdoor conditions where consistent illumination cannot be guaranteed [28]. In our case, this method helps to isolate brightness variations that are indicative of Fusarium infection, which might otherwise be obscured in the RGB space due to color shifts.

We used the cv2.cvtColor function from OpenCV [52] to convert the ROI from RGB to the HSV color space. Once the images are converted to the HSV color space, we apply a threshold to the Value channel (V) to isolate bright pixels, which are presumed to represent the Fusarium-infected areas. The thresholding operation is defined as:

$$V_{\text{thresholded}} = \begin{cases} 255 & \text{if } V > T, \\ 0 & \text{if } V \leq T, \end{cases} \tag{3}$$

where V is the pixel intensity in the Value channel, and T is the threshold value. Unlike prior work that often uses fixed or purely empirically selected thresholds, we compute T in a scalable, data-driven manner. Specifically, T is derived from the mean intensity of randomly selected Fusarium-infected regions across 30 full-resolution images (4000 × 3000 pixels). This approach accounts for variability in leaf brightness, lighting conditions, and disease expression across images, improving the robustness and generalizability of FISI estimation.

To estimate T, a histogram is generated from these selected regions, and the mean value of the histogram is calculated. This mean value is used as the threshold T.

Mathematically, the threshold T is computed as:

$$T = \frac{1}{N} \sum_{i=1}^{N} V_i \tag{4}$$

where N is the total number of pixels belonging to detected Fusarium regions, and V_i represents intensities across the individual pixels i in the Value channel.

After thresholding, we count the number of bright pixels (N_{bright}) and calculate the total number of pixels in the region (N_{total}). The severity of the disease is quantified as the percentage of bright pixels:

$$\text{FISI} = \frac{N_{\text{bright}}}{N_{\text{total}}} \times 100 \tag{5}$$

Where:

- N_{bright} is the number of bright pixels (those with $V > T$),
- N_{total} is the total number of pixels in the ROI.

After quantifying Fusarium wilt severity as a percentage, labels are assigned based on the following thresholds: $0 \leq \text{value} < 40$ for Category 1, $40 \leq \text{value} < 55$ for Category 2, $55 \leq \text{value} < 65$ for Category 3, and $65 \leq \text{value} \leq 100$ for Category 4. This adaptive thresholding extends previous pixel-brightness-based disease severity methods by refining the threshold selection for each image, which reduces bias and improves accuracy under varying illumination conditions.

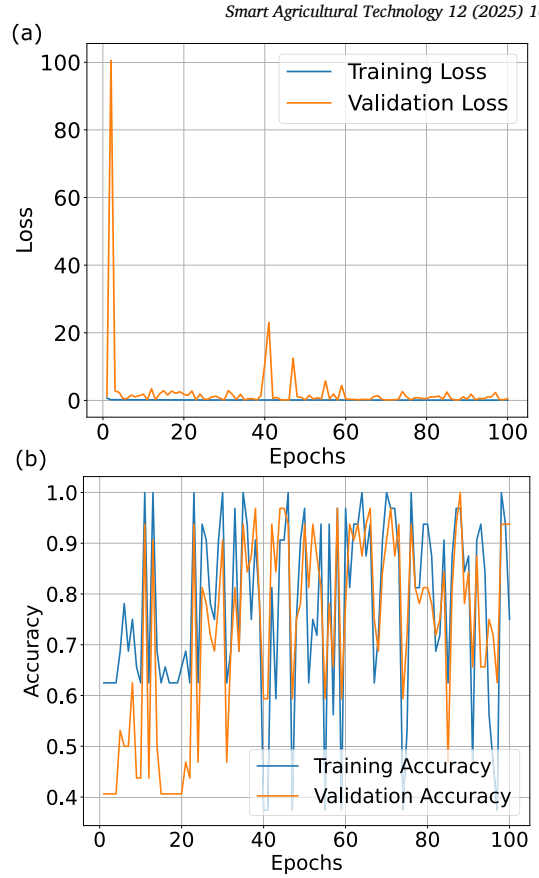

Fig. 5. Training Performance: (a) Loss vs Epochs and (b) Accuracy vs Epochs for the Proposed Customized ResNet Model.

4. Experimental results

4.1. Performance analysis of Fusarium wilt detection

4.1.1. Candidate Fusarium wilt detection results

We identified candidate Fusarium-infected regions by detecting the specific light tan color range. Overall, we visually identified 68 candidate Fusarium-infected areas across 10 image patches. Of these 68 candidate areas, 63 were correctly identified by our model, while 5 were misclassified. Additionally, 40 non-Fusarium regions were incorrectly marked as candidate Fusarium-infected, reflecting the model's sensitivity but also indicating the need for further validation. This tan-based detection represents an intermediate step in our two-stage Fusarium wilt detection process, designed to ensure that all potentially infected regions are captured for further analysis. The subsequent application of the proposed LeafyResNet model refines these candidate regions, effectively reducing false positives and confirming actual Fusarium infections. The initial misclassifications and false positives mainly occurred due to the visual similarity between Fusarium symptoms and surrounding soil or stem areas with similar tan intensities. Despite these challenges, the model showed strong performance with an overall precision of 92.65%. In severe Fusarium cases, where the infection color was very close to the soil color, the model struggled to distinguish them, leading to missed detections. However, excluding these severe cases, the precision significantly improved, surpassing 99%. This highlights the model's ability to accurately identify infections in early-stage Fusarium infections.

In Fig. 6 (a-b), we visualize the intermediate candidate Fusarium wilt detection results, where the identified regions are highlighted by rectangles. These "candidate regions" represent areas that may show Fusarium-like discoloration, but do not confirm the presence of Fusar-

Fig. 6. Workflow of Fusarium Wilt Detection and Severity Quantification in Lettuce. (a) Original Image (b) candidate Fusarium Wilt Detection (c) Confirm Fusarium Wilt, and (d) Quantification of Severity in detected regions, with each detected region labeled by a number followed by its severity in parentheses, e.g., 1 (4) indicating detected region 1 with severity level 4.

Table 3
Ablation study of LeafyResNet with different hyperparameter settings.

Filter	Learning Rate	Batch Size	Accuracy	Data Augmentation
64-128-256-512	0.001	32	96.3%	Yes
64-128-256-512	0.005	32	96.3%	Yes
64-128-256-512	0.0005	32	95%	Yes
64-128-256-512	0.001	32	91.1%	No
64-128-256-512	0.005	32	95%	No
64-128-256-512	0.0005	32	91.1%	No
128-256-512-1024	0.001	32	97.04%	Yes
128-256-512-1024	0.005	32	95.56%	Yes
128-256-512-1024	0.0005	32	94.07%	Yes
64-128-256-512	0.001	64	93.3%	Yes
64-128-256-512	0.005	64	94%	Yes
64-128-256-512	0.0005	64	88.1%	Yes

ium wilt. Further confirmation is required, which is addressed in the next section using the proposed LeafyResNet model.

The proposed LeafyResNet model demonstrated strong performance in confirming Fusarium infections within the candidate areas. The training performance of the proposed model is shown in Fig. 5, where (a) illustrates the loss curve over epochs, and (b) displays the accuracy progression during training. These plots provide insights into how the model converges over time, highlighting its efficiency in both, minimizing loss and maximizing accuracy. Evaluation metrics on the test set for the proposed LeafyResNet model are summarized in Table 4, showing an overall accuracy of 96.3%, precision of 94.1%, recall of 100%, F1-score of 97.1%, mAP of 99.5%, along with a False Positive Rate (FPR) of 4%. Fig. 6 (b) illustrates the candidate Fusarium-infected regions detected using light tan value evaluation, highlighted by rectangles. These regions were identified based on light tan discoloration intensity but included false positives due to visual similarities with infected areas, which caused confusion in the detection process. Fig. 6 (c) shows the confirmed Fusarium-infected areas after applying the LeafyResNet model. The model effectively removed these false positives, retaining only the true infections. This two-step process improved accuracy by confirming the initially detected Fusarium infections.

4.1.2. Ablation study on LeafyResNet hyperparameters

To evaluate the impact of hyperparameter choices on LeafyResNet performance, we conducted an ablation study by varying filter configurations, learning rates, and batch sizes. All ablation experiments were performed using 40 epochs, which were iteratively determined for best runtime and quality performance.

As shown in Table 3, in the first set of experiments with filter sizes 64-128-256-512, a learning rate of 0.001, and a batch size of 32, the model achieved 96.3% accuracy. This was the same as the accuracy achieved with a slightly higher learning rate of 0.005. However, when the learning rate was lowered to 0.0005, accuracy dropped to 95%. As mentioned earlier, data augmentation was applied only to the training data, while the testing data remained unchanged to ensure fairness.

When the same settings were used without data augmentation, performance dropped to 91.1% with a learning rate of 0.001. This shows that data augmentation significantly impacts performance, which is why the remaining combinations were evaluated with augmentation. Increasing the filter configuration to 128-256-512-1024 improved accuracy to 97.04% with a learning rate of 0.001, but this came at the cost of a higher false positive rate (FPR) of 6.57% and increased computational cost. Other combinations, such as using higher or lower learning rates with the larger filter configuration, resulted in lower accuracy (95.56% and 94.07%).

Furthermore, experiments with a larger batch size of 64 generally led to lower accuracy across all learning rates, with the best result being 94% (using filters 64-128-256-512 and a learning rate of 0.005). Therefore, the final LeafyResNet model uses the filter configuration 64-128-256-512, a learning rate of 0.001, and a batch size of 32 with data augmentation for balanced performance.

4.1.3. Performance comparison with state-of-the-art models

To evaluate the effectiveness of the proposed model, we compared its performance with state-of-the-art deep learning models using the proposed lettuce dataset. The first benchmark model, Inception-V3 [9], was originally designed to distinguish between healthy and Fusarium-infected regions in radish. The second model, VGG-16 [22], was previously used to classify healthy and wilted radish plants. We implemented these models on our lettuce dataset for a direct comparison. Additionally, we included two ResNet variants (ResNet18 [24] and ResNet34 [53]), to further analyze model performance.

A comprehensive performance comparison was conducted between the proposed LeafyResNet model, Inception-V3, VGG-16, and the two ResNet variants (ResNet18 and ResNet34). LeafyResNet inference and performance tests were conducted on a 13th Gen Intel(R) Core(TM) i5-

Table 4
Performance and cost analysis of different models on the lettuce dataset.

Model	Accuracy	Precision	Recall	F1-Score	FPR	mAP	GFLOPs	Training Time (s)
LeafyResNet	**96.3%**	94.1%	**100%**	**97.1%**	**4%**	99.5%	4.72	6.34×10^3
ResNet18	95.0%	**96.1%**	95.1%	95.5%	5%	99.6%	3.72	4.58×10^3
ResNet34	96.0%	94.0%	99.0%	96.3%	5%	**99.7%**	6.72	8.86×10^3
Inception_v3	70.4%	69.0%	88.5%	77.5%	33%	70%	**0.74**	6.43×10^2
VGG16	89.0%	87.1%	95.1%	91.1%	12%	96%	5.94	$\mathbf{2.43} \times 10^2$

1340P processor (1.90 GHz) with 16 GB RAM (15.7 GB usable, 64-bit OS, x64-based processor). All models were trained on the same lettuce dataset to ensure a fair evaluation. The results are summarized in Table 4. The best results for each metric are highlighted in bold for clarity.

The proposed LeafyResNet model outperforms the comparison models in terms of overall performance, achieving 96.3% accuracy, 94.1% precision, 100% recall, 97.1% F1-score, a low false positive rate (FPR) of 4%, and an mAP of 99.5%. Compared to the best baseline model, ResNet34, LeafyResNet shows a 0.3% improvement in accuracy (96.3% vs. 96.0%), a 0.8% improvement in F1-score (97.1% vs. 96.3%), and a reduction in FPR from 5% to 4%, as a measurable improvement over the state-of-the-art. Although ResNet18 achieved a slightly higher precision of 96.1%, LeafyResNet excels in overall accuracy, recall, F1-score, and FPR reduction.

In addition to performance metrics, a cost-performance analysis was performed using GFLOPs (Giga Floating Point Operations), training time, and inference time. GFLOPs indicate the computational complexity of the model, while inference time was measured over 135 samples to evaluate practical deployment efficiency. LeafyResNet requires 4.72 GFLOPs, 6.34×10^3 seconds for training, and 9.90 seconds for inference. Notably, its inference time is slightly faster than ResNet18 (9.99 s) paired with higher overall accuracy, resulting in better efficiency during deployment. Compared to ResNet34, LeafyResNet achieves higher accuracy and F1-score with lower GFLOPs, shorter training time (6.34×10^3 s vs. 8.86×10^3 s), and faster inference (9.90 s vs. 11.8 s). Inception_v3 and VGG16 either have low computational cost but poor accuracy. Overall, LeafyResNet achieves the best trade-off between performance and computational cost, for Fusarium wilt detection on the lettuce UAV dataset.

4.2. Performance evaluation of FISI

The model was further tested for its ability to quantify the severity of Fusarium infections in lettuce crops. By analyzing the pixel composition within the confirmed infected regions, we assessed the severity of the infection. Fusarium wilt severity was quantified by assigning labels (0 for healthy, 1–4 for varying severity) in the detected Fusarium wilt regions, assigning non-detected areas as 0. The quantification process evaluated the distribution of labels within the identified regions, with higher label values representing more severe infections. For example, the confirmed regions in Fig. 6 (d) show nine detected areas, each enclosed in a bounding box and labeled with its box number followed by its severity level in parentheses (e.g., 1(2), 2(4)), demonstrating the model's ability to distinguish different infection severity levels. To evaluate the model we used severity data collected manually in the south field. Ground truth labels (0–4) were assigned to each plant by experts at the Yuma Center of Excellence for Desert Agriculture. While the model was tested on the entire south field dataset, 10 randomly selected patches containing 112 plants were specifically used for quantitative evaluation. The model achieved an overall accuracy of 86%, performing best for Class 0 (96.3% accuracy, 50 plants). For infected plants, accuracy was 80% for Class 1 (10 plants), while Classes 2–4 ranged from 69% to 76% accuracy across 67 plants. In our experience, temporal differences between UAV image collection and expert disease scoring posed challenges in accurately matching plants. To address this, we carefully adjusted the assignments and visually rescored the plants to ensure accuracy. The model demonstrated reliability to identify FISI, despite some overlap in the classification of severity stages, particularly between Class 3 and Class 4, and Class 2 and Class 3. The consistency between FISI and expert ratings was also quantified using Cohen's Kappa coefficient, resulting in a value of 80.20%, indicating substantial agreement beyond random agreement.

4.3. Discussion

The proposed method for detecting Fusarium wilt and scoring Fusarium infection severity demonstrated improved performance compared to state-of-the-art methods but also encountered challenges. By integrating RGB thresholding for initial detection with the deep learning model LeafyResNet, we achieved high accuracy (96.3%), precision (94.1%), recall (100%), and an F1-score of 97.1%, with a false positive rate of 4%. However, detection accuracy declined particularly at advanced infection stages, when the color of Fusarium-infected areas became similar to that of the soil. However, emphasizing the light tan color during model training alleviated some of the previously reported limitations (e.g., [8]), although some detection challenges remain when color differences become minimal. These challenges arise when severely infected Fusarium lightens, causing pixel values to decrease and become too similar to the soil average color range (216, 200, 153). To address this limitation, future improvements could include incorporating hyperspectral imaging to enhance contrast, utilizing texture-based features to better distinguish infected regions, expanding the training set with additional examples of severely infected plants, and employing adaptive thresholding or temporal information to track disease progression. Implementing these strategies in conjunction with the current color-based intermediate step may help mitigate performance degradation and enhance detection across both early and advanced stages of Fusarium infection.

It is also important to note that Fusarium wilt can resemble other wilting conditions, such as those caused by water stress or root damage due to insect feeding. While our evaluations were conducted in a field where Fusarium wilt was the only disease present, the presence of other pests or diseases in the field could lead to misidentifying other diseases as Fusarium wilt. Incorporating these additional diseases into the training data could help distinguish Fusarium wilt from other similar conditions.

Notably, our model excelled in early stage detection, achieving a precision of 92.65%, which increased to over 99% when severe cases were excluded. Comparative analysis with state-of-the-art models, including Inception_v3, VGG16, ResNet18, and ResNet34, confirmed the superior performance of our proposed LeafyResNet model in detecting Fusarium wilt in different datasets. Future improvements could integrate environmental metadata to enhance model robustness.

The severity quantification of our model achieved an overall accuracy of 86% against expert-labeled ground truth data from the Yuma Center of Excellence for Desert Agriculture. However, challenges in plant-to-plant matching due to time differences between UAV image collection and visual disease scoring affected classification accuracy. Careful adjustments during validation helped mitigate these discrepancies, ensuring reliable severity assessments. Expanding this approach to diverse crops and integrating real-time drone-based detection could further enhance its applicability in precision agriculture, improving disease monitoring and management strategies.

From a scientific standpoint, our method also contributes to advancements in root biology. Fusarium infection originates at the root, and

root architecture is one potential indicator of a plant's tolerance to biotic stressors, such as the parasitic weed Striga, particularly in crops like cowpea [6] and sorghum [32,31]. While a direct pathogenic comparison between Striga and the soil-borne fungus Fusarium is not feasible due to their differing infection mechanisms, both interact directly with the root system. This suggests that phenotyping root architectural traits could play a pivotal role in understanding and potentially enhancing plant tolerance to Fusarium infection using modern 3D root phenotyping tools [42,43] by associating Fusarium severity and root architecture phenotype.

5. Conclusion

We have developed a highly effective method for detecting Fusarium wilt in lettuce and accurately estimating the FISI using UAV imagery. By integrating light tan evaluation for initial detection with a deep learning model featuring the LeafyResNet architecture, our approach delivers exceptional performance, achieving test results of 96.3% accuracy, 94.1% precision, 100% recall, and a 97.1% F1-score, with a false positive rate of just 4%, outperforming the state-of-the-art models, including Inception_v3, VGG16, ResNet18, and ResNet34. Moreover, our model demonstrated an 86% overall accuracy in assessing Fusarium infection severity, which was validated against expert-annotated ground truth.

Critically, our approach excels at identifying early-stage symptoms of Fusarium wilt in lettuce, enabling timely intervention to mitigate crop losses. By detecting infections at different stages of progression, our model not only enhances disease monitoring but also provides precise quantification of infection severity across affected regions. These capabilities make it a promising tool for precision agriculture and can be integrated into state-of-the-art data processing pipelines used for lettuce [19]. Therefore, the presented method offers growers reliable, data-driven insights to optimize disease management strategies.

Looking ahead, expanding this methodology to other crops and plant diseases will enhance its applicability across diverse agricultural landscapes. Further advancements in real-time, drone-based detection will empower farmers with instant, actionable intelligence to reduce yield losses, minimize chemical inputs, and promote sustainable farming practices. However, limitations remain, including reduced accuracy in severely infected regions, and generalizability to other crops remains to be tested. Despite these limitations, the approach enables early detection of infections, supporting precision agriculture by potentially reducing yield losses and minimizing unnecessary chemical applications.

Additionally, refining our quantification approach to assess infection severity across entire fields—rather than isolated detected regions—will provide a more comprehensive evaluation of disease impact. Such advancements will enable more accurate yield loss estimations and economic impact assessments, further strengthening decision-making in precision agriculture.

In conclusion, our model's performance underscores its transformative potential for real-world agricultural applications. Future work will focus on improving detection accuracy for advanced infection stages and integrating multispectral or texture-based features to better distinguish subtle symptoms. Future research will also refine LeafyResNet's ability to distinguish between adjacent severity levels, further enhancing its precision and effectiveness. With its proven capability in early detection and severity assessment, this approach represents a significant step forward in disease monitoring and crop protection.

CRediT authorship contribution statement

Kabir Hossain: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Stephanie Slinski:** Writing – review & editing, Resources, Investigation, Formal analysis, Data curation. **Alexander Bucksch:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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